

最小生成树

《算法设计与分析》

信息科学与技术学院 张德富





最小生成树

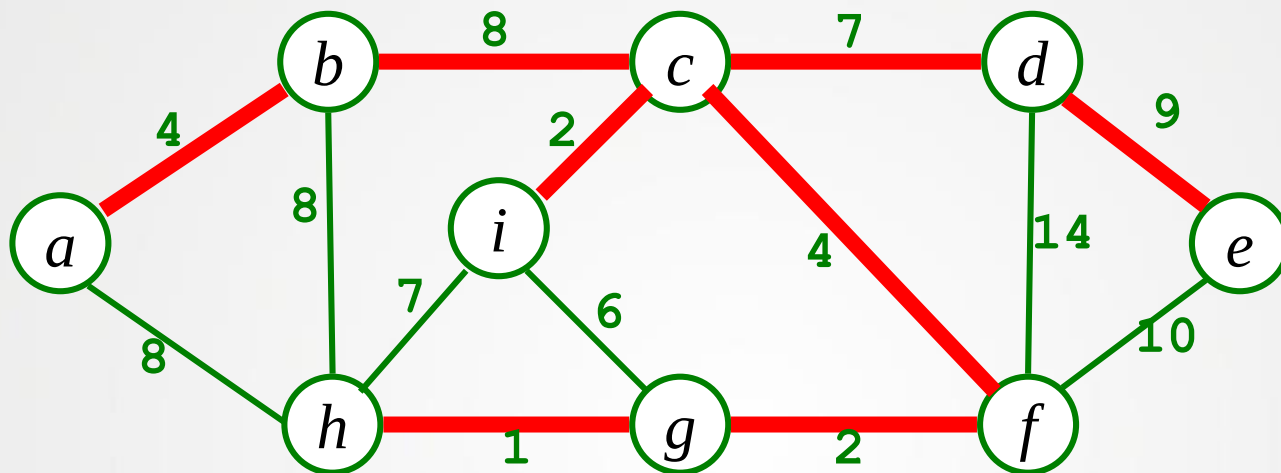
- 最小生成树(Minimum Spanning Trees, MST)问题可以描述如下:
给定一无向连通图 $G = (V, E)$, 对每条边 $(u, v) \in E$, 都有一相应的权值 $w(u, v)$, 表示连通 u 和 v 的代价。我们希望找到一个无环子集 $T \subseteq E$, 其连通了图所有的顶点且总权值

$$w(T) = \sum_{(u,v) \in T} w(u, v)$$

最小。



最小生成树



- 两个贪心算法来求解最小生成树问题:

Kruskal 和 Prim 算法

- 贪心策略每次选择一条边,逐步生成最小生成树。



最小生成树

Kruskal 和 Prim 算法

- **Kruskal算法是一个贪心算法，它每一次从剩余的边中选一条权值最小的边，然后加入到一个边的集合。因为每一步其加到森林中的都是权值最小的边。这条边要么连接两棵树，要么加到一棵树上，要么生成一棵新树。当所有的顶点都连接后停止。**

MSTKruskal(G, w)

1 $A \leftarrow \emptyset$

2 **for** each vertex $v \in V$ **do**

3 MakeSet(v)

4 sort the edges of E by weight w

5 **for** each edge $(u, v) \in E$ **do**

6 **if** FindSet(u) $\neq$ FindSet(v) **then**

7 $A \leftarrow A \cup \{(u, v)\}$

8 Union(u, v)

9 **return** A

$O(|E| \lg |V|)$

} $O(|V|)$

→ $O(|E| \lg |E|)$

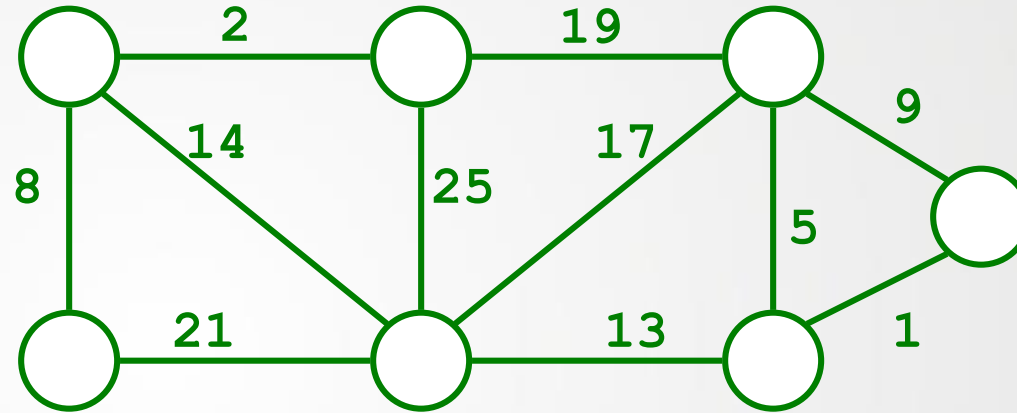
→ $O(|E|)$

} $O(\lg |V|)$

An example

MSTKruskal(G, w)

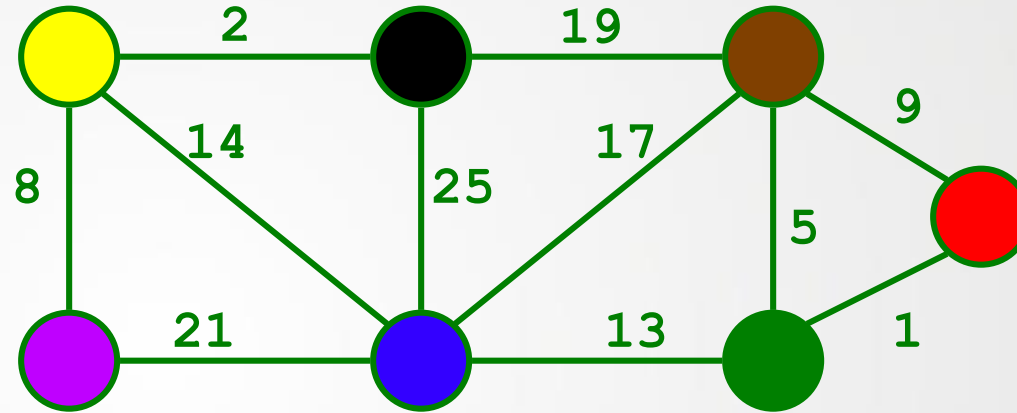
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8     Union( $u, v$ )
9 return  $A$ 
```



An example

MSTKruskal(G, w)

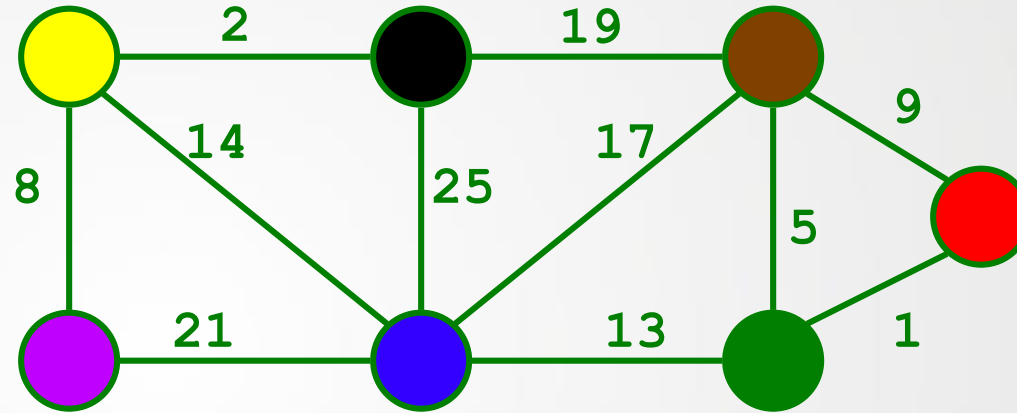
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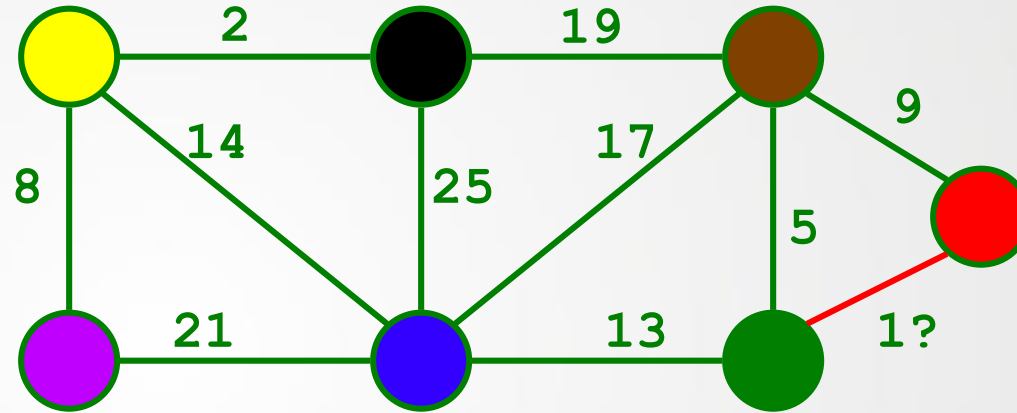
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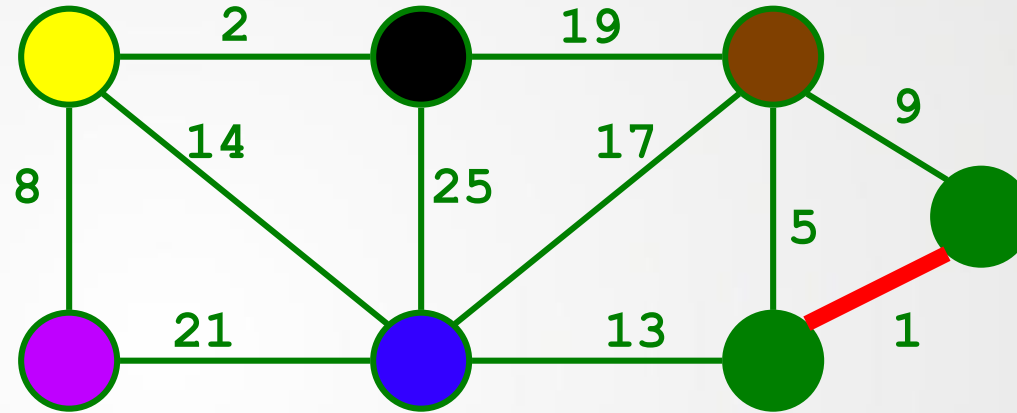
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An example

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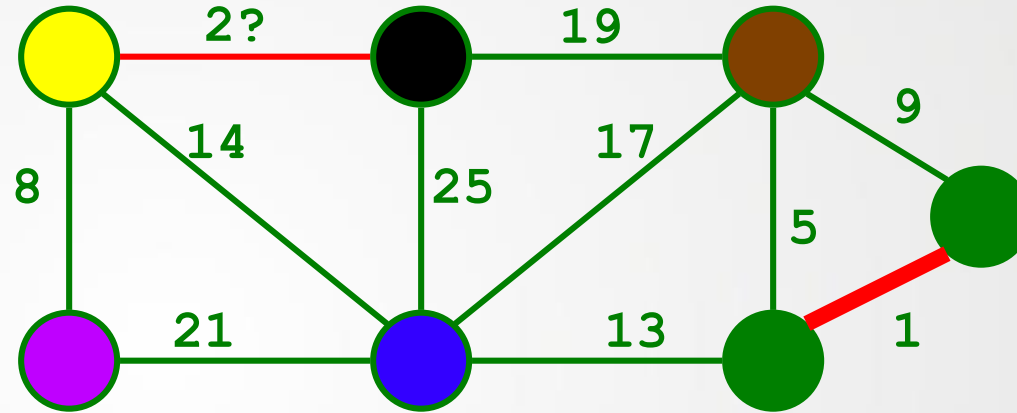
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An example

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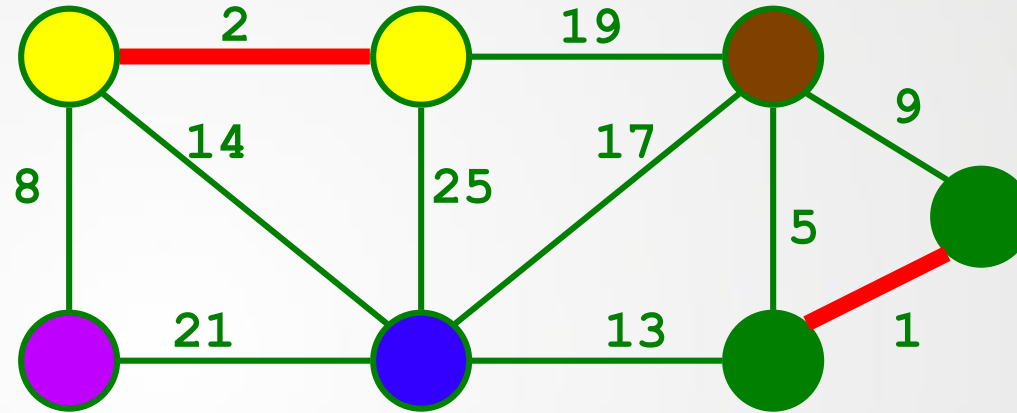
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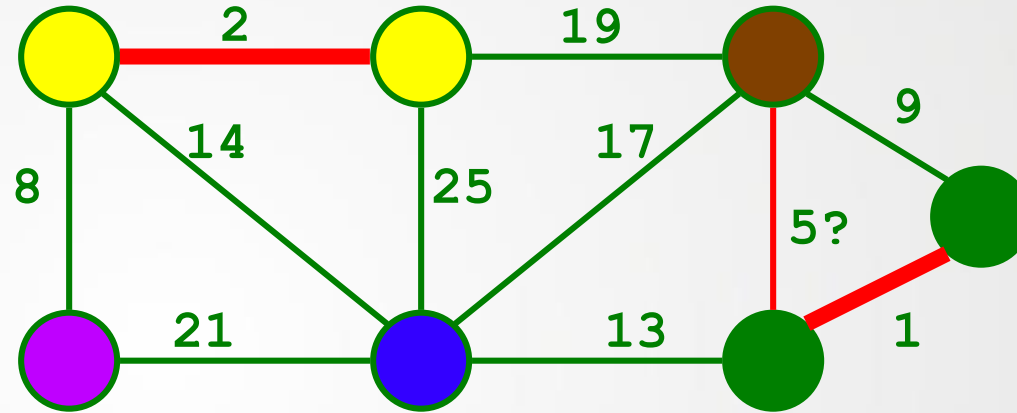
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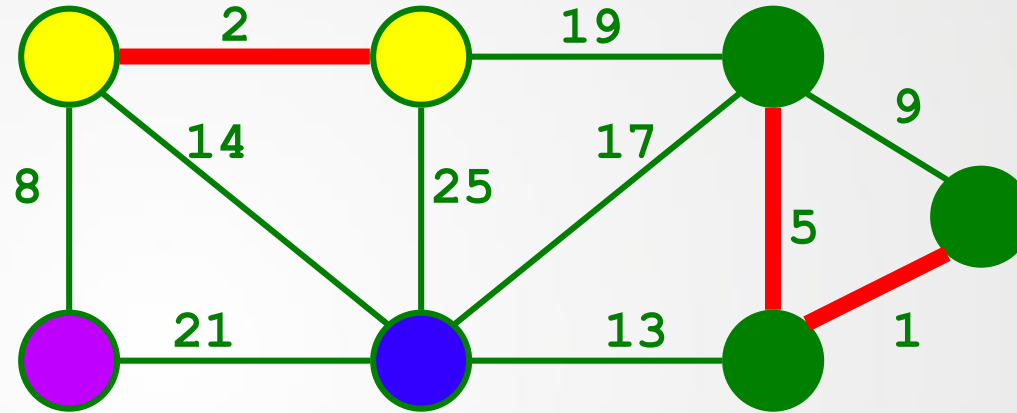
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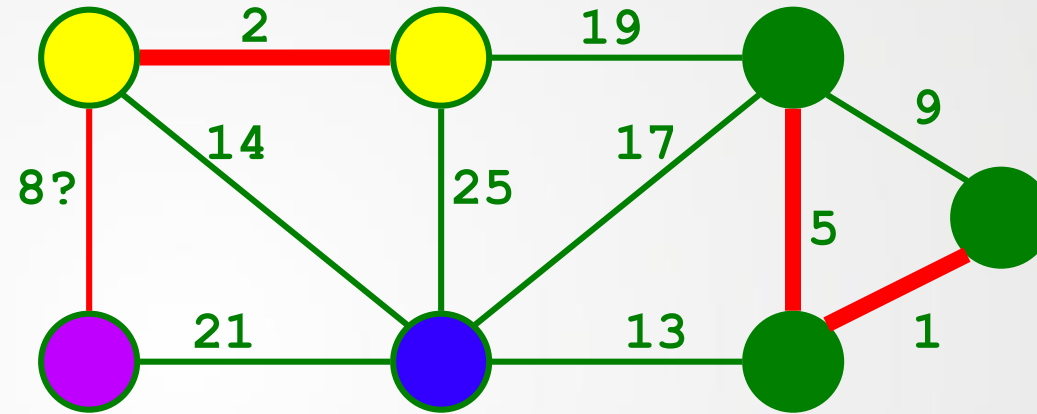
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An example

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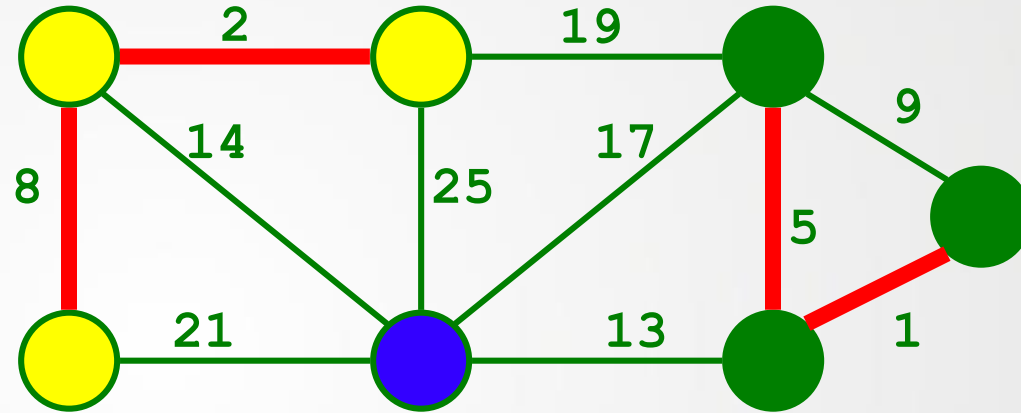
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An example

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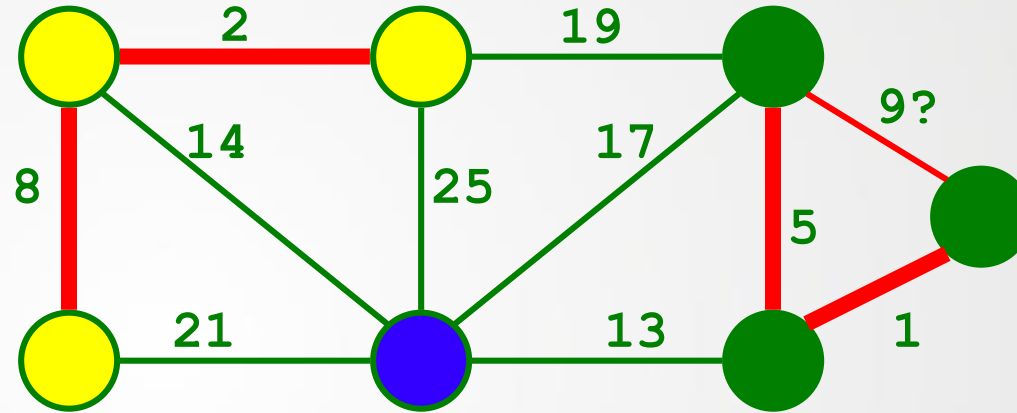
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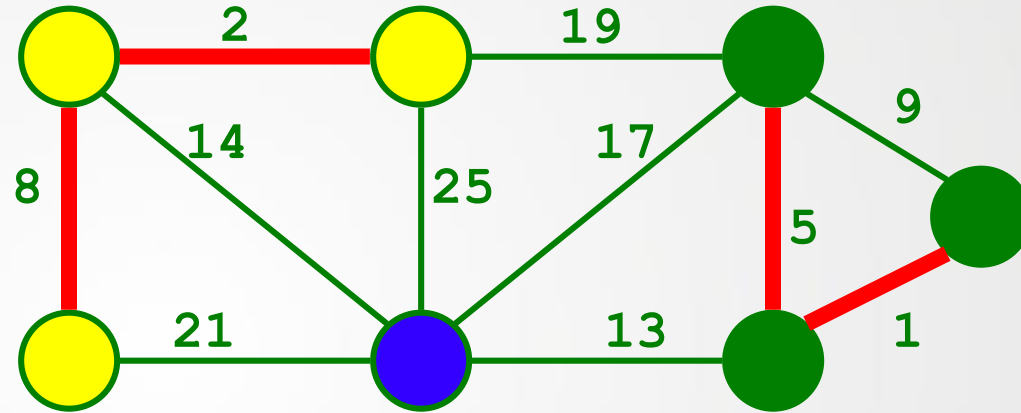
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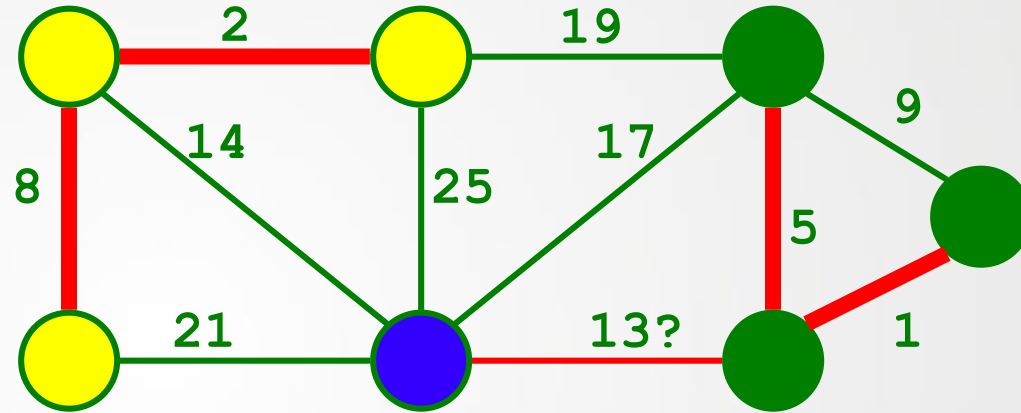
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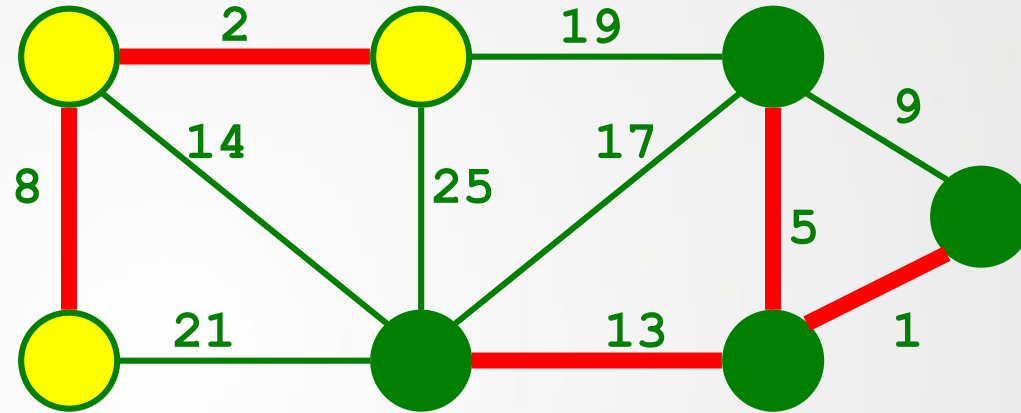
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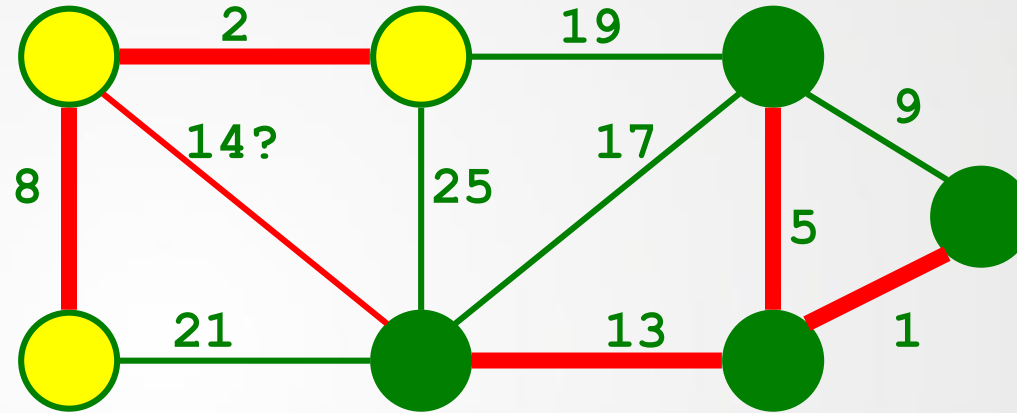
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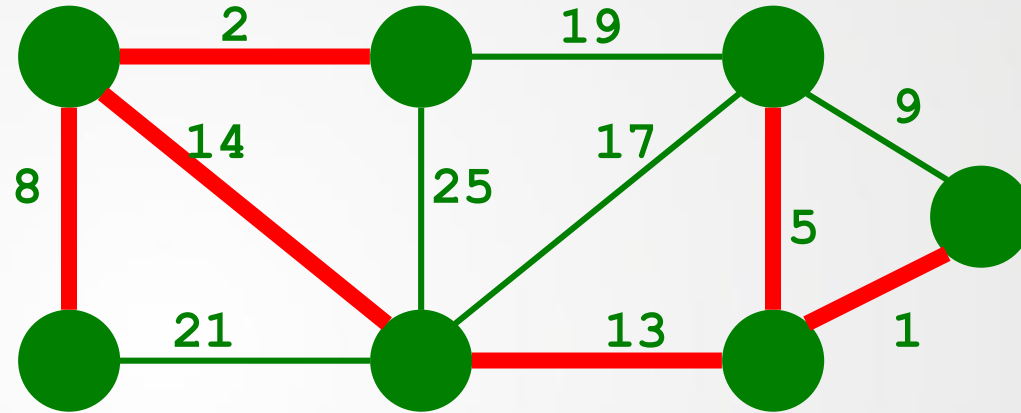
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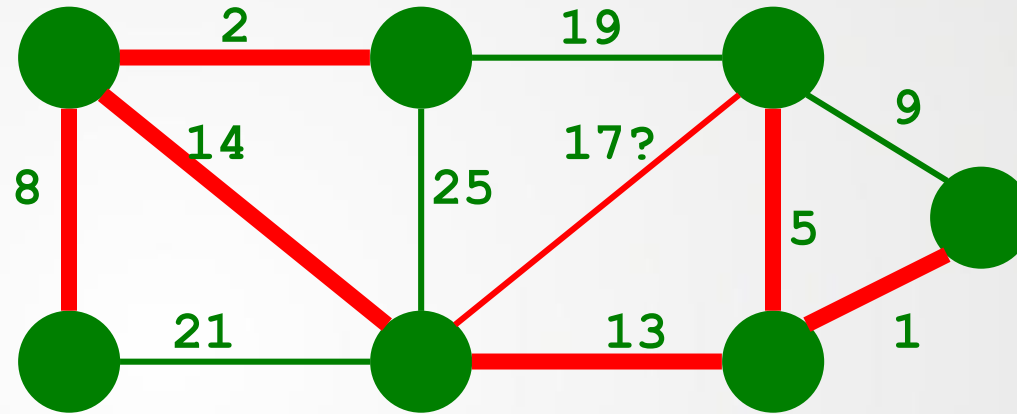
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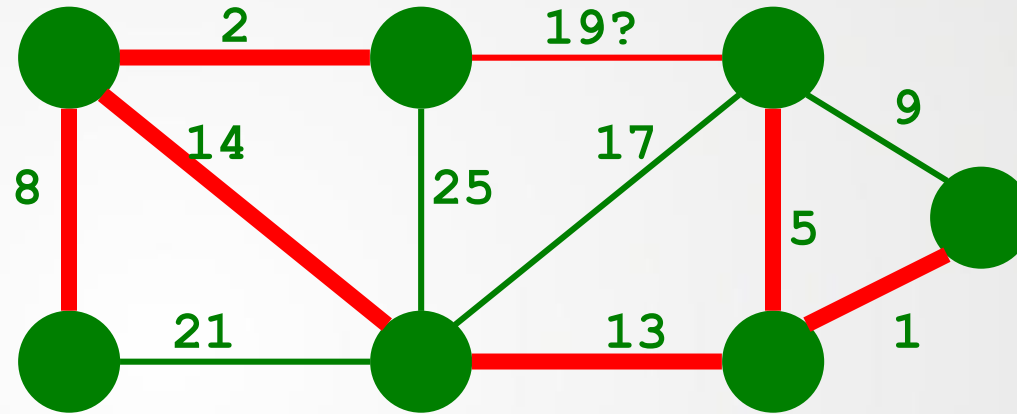
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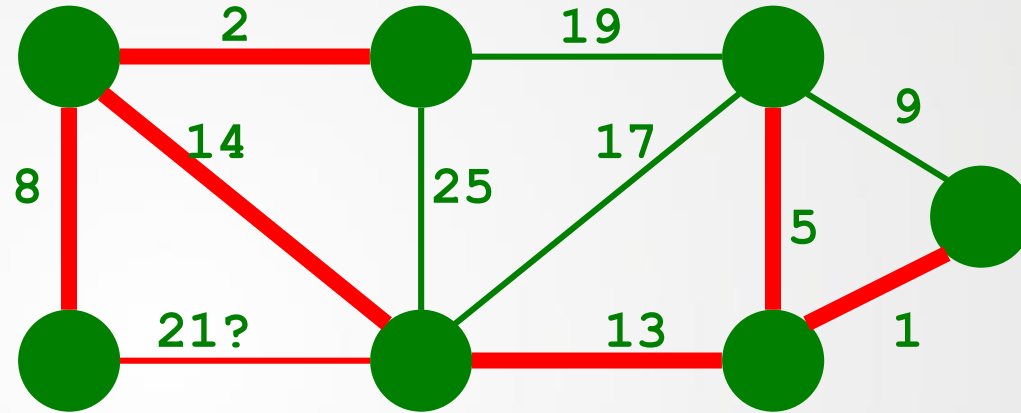
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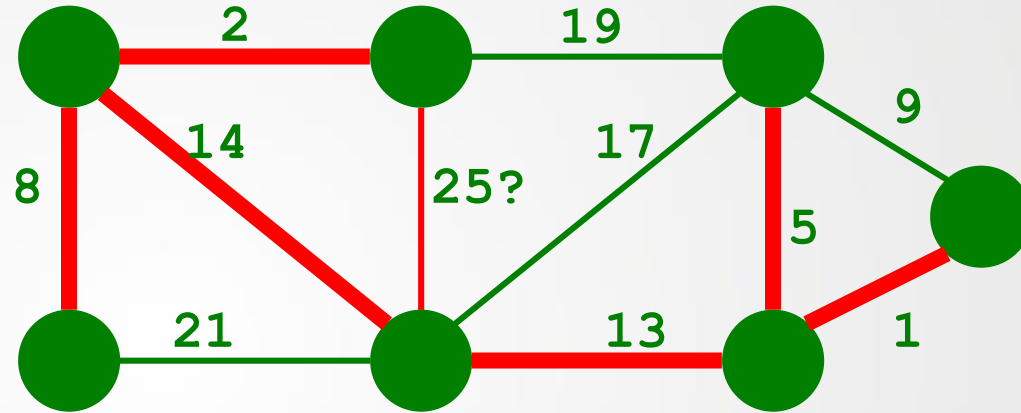
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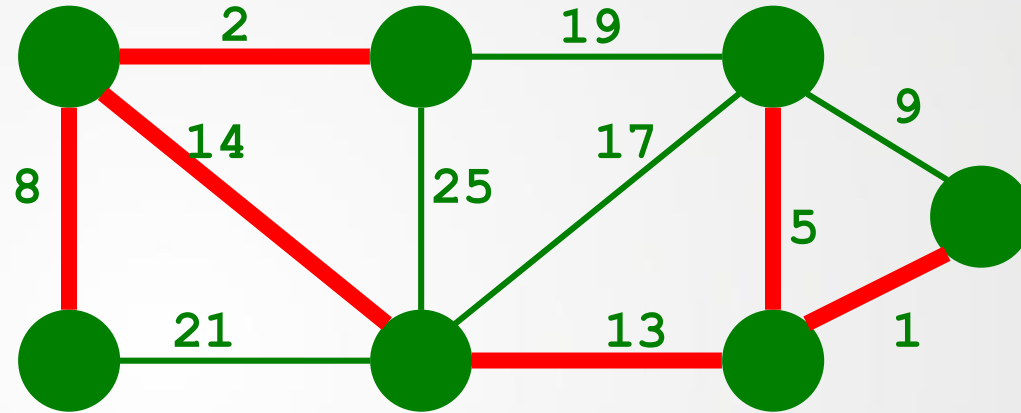
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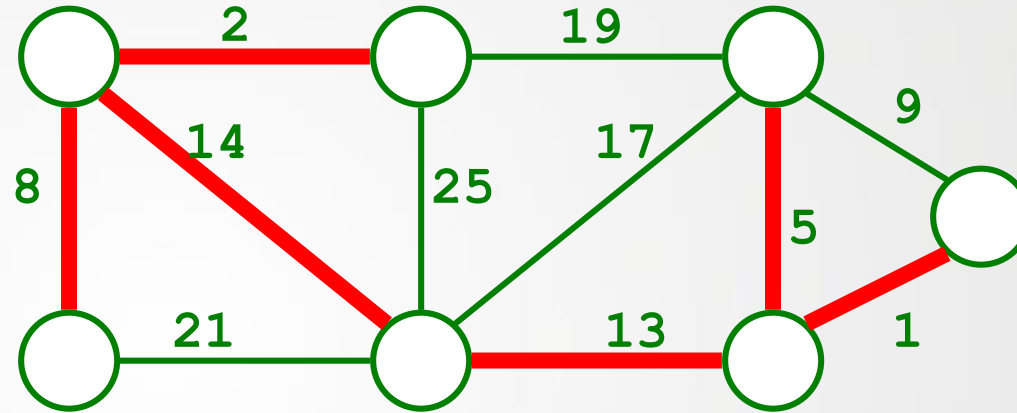
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最小生成树

定理8.4：

- 任给带权连通无向图，KruskalMST算法都能找到该图的一棵最小生成树。
- 证明：为了证明这个结论，我们只需证明存在下述**循环不变量**：
在每次往集合 A 中增加一条边时，集合 A 均是一棵最小生成树边集的一个子集。令 T 为一棵最小生成树， A^* 是 T 的边集，则只需证明 $A \subset A^*$ ， A 是 A^* 的一个子集即可，即。也就是说每一步贪心选择的边都在最小生成树中。

初始步：第1行结束后，集合 $A = \emptyset$ ，显然是 A^* 的子集，循环不变量为真；

归纳步：假设往 A 中加入边 (u,v) 前， A 是最小生成树边集的一个子集，令 $A' = A \cup (u,v)$ ，我们只需证明 A' 也是 A^* 的一个子集即可。根据归纳假设 $A \subset A^*$ 。如果 A^* 包含 (u,v) ，则结论显然成立。如果 A^* 不包含 (u,v) ，由于 T 中顶点是连通的，则 $A^* \cup (u,v)$ 刚好包含一个回路， (u,v) 是其中的一条边，而且是当前选择要加入的边。令 X 是当前包含顶点 u 的子树的顶点集，因为 (u,v) 连接 X 中的一顶点 u 与 $V-X$ 中的另一顶点 v ， A^* 必须包含另外一条边 (x,y) 使得 $x \in X$ 和 $y \in V-X$ 且该边不在KruskalMST算法所得到的生成树中，这样的边一定能够找到，否则 A^* 的边会全部在KruskalMST算法所得到的生成树中。现在考虑边 (u,v) 和 (x,y) 权值的大小。由于 (x,y) 不在 A 中，故一定有 $w(u,v) \leq w(x,y)$ ，否则，KruskalMST算法将会选择边 (x,y) ，与此时它选 (u,v) 矛盾。令 $A^{**} = \{A^* - (x,y)\} \cup (u,v)$ ，那么 A^{**} 也是一棵最小生成树。由于 $A' \subset A^{**}$ ，因此， A' 也是最小生成树边集的一个子集。

终止步：当KruskalMST算法结束时，所有被加到中的边都在最小生成树中，所以第9行返回的必为最小生成树。



最小生成树

Prim 算法

- Prim算法的特点是集合的所有的边都是来自一棵单一的树，这棵树开始于任意一个根顶点，然后逐步生长，直到能包含所有中的顶点为止。

MSTPrim(G, w, r)

1 **for** each $u \in V$ **do**

2 $key[u] \leftarrow \infty$

3 $\pi[u] \leftarrow \text{NIL}$

4 $key[r] \leftarrow 0$

5 $Q \leftarrow V$

6 **while** $Q \neq \emptyset$ **do**

7 $u \leftarrow \text{ExtractMIN}(Q)$

8 **for** each $v \in \text{Adj}[u]$ **do**

9 **if** $v \in Q$ and $w(u, v) < key[v]$ **then**

10 $\pi[v] \leftarrow u$

11 $key[v] \leftarrow w(u, v)$

12 DecreaseKey(Q, v, key)

$O(|V|)$

$\longrightarrow O(\lg|V|)$

$O(\lg|V|)$

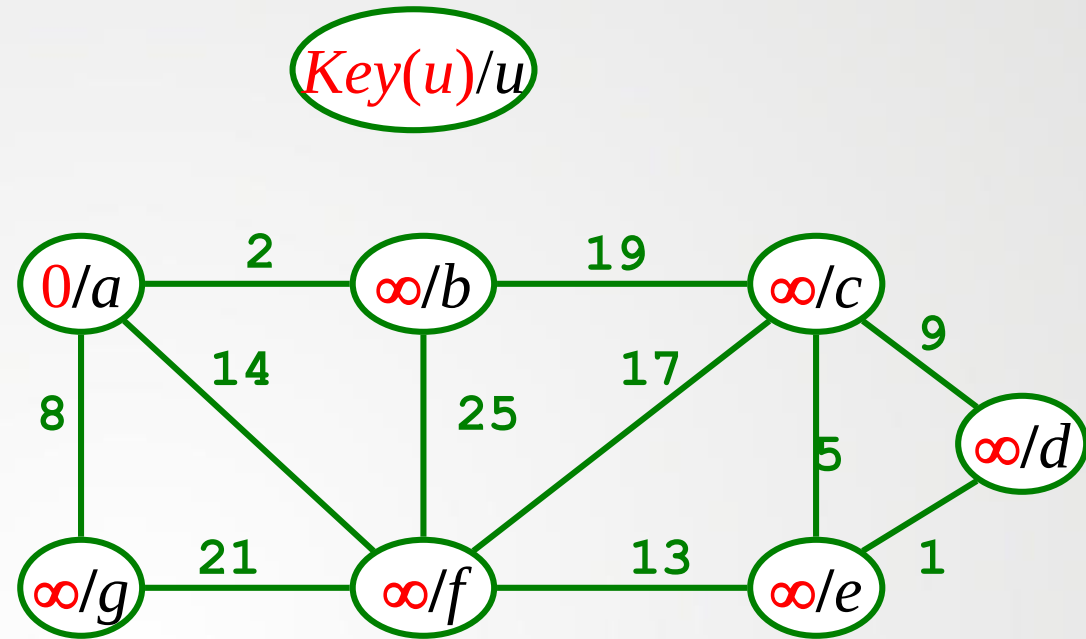
$O(|E|)$

$O(|E|(\lg|V|+|E|))$

An example

MSTPrim(G, w, r)

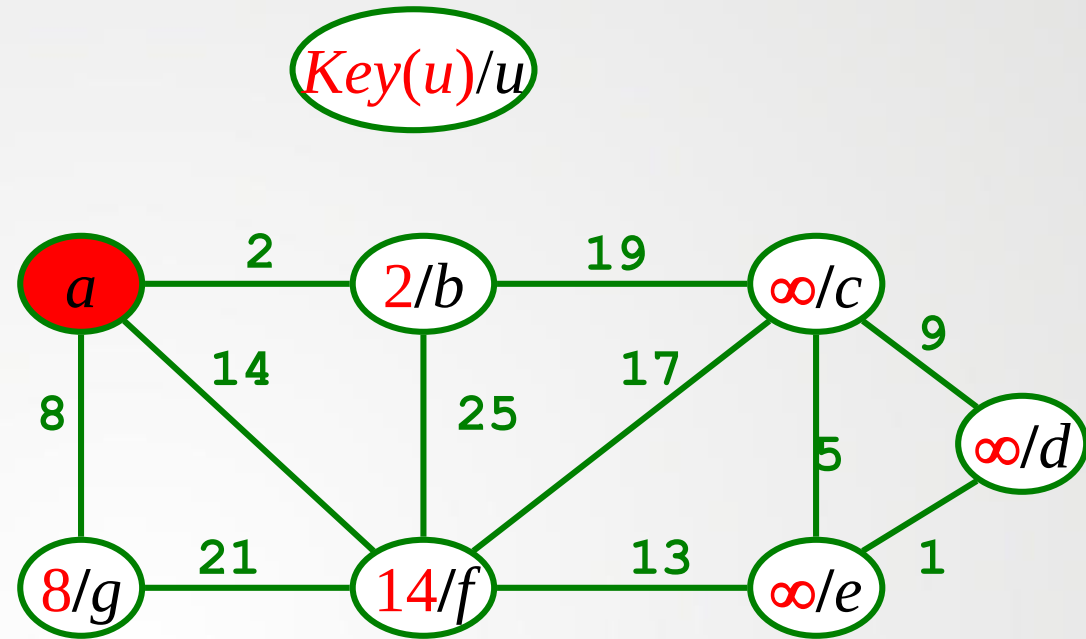
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- 8 **for** each $v \in \text{Adj}[u]$ **do**
- 9 **if** $v \in Q$ and $w(u, v) < key[v]$ **then**
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An example

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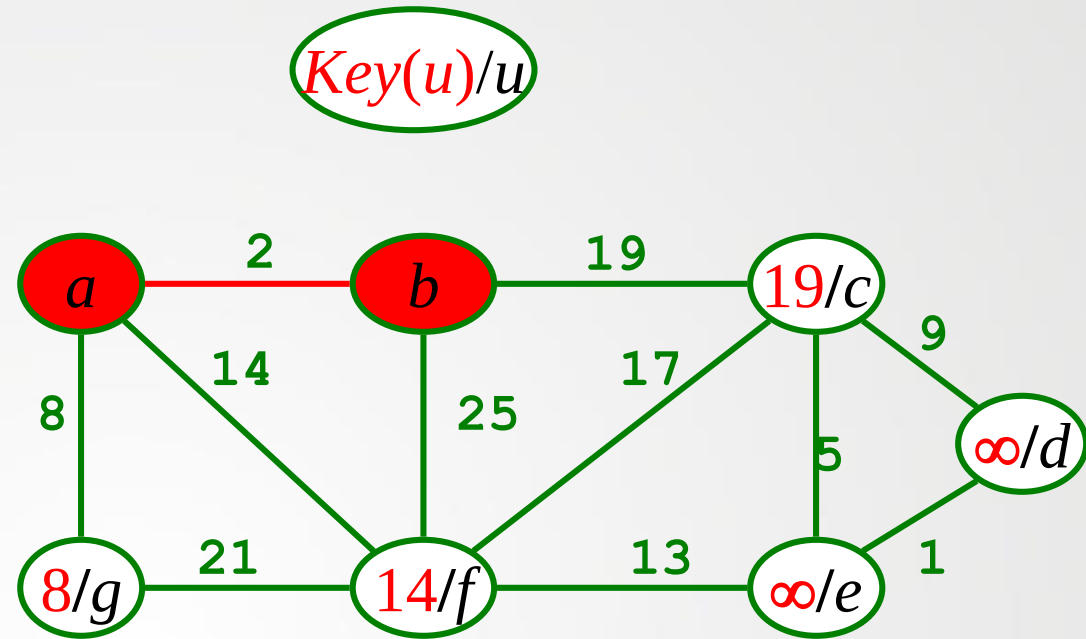
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- 10 $\pi[v] \leftarrow u$
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- 12 DecreaseKey(Q, v, key)



An example

MSTPrim(G, w, r)

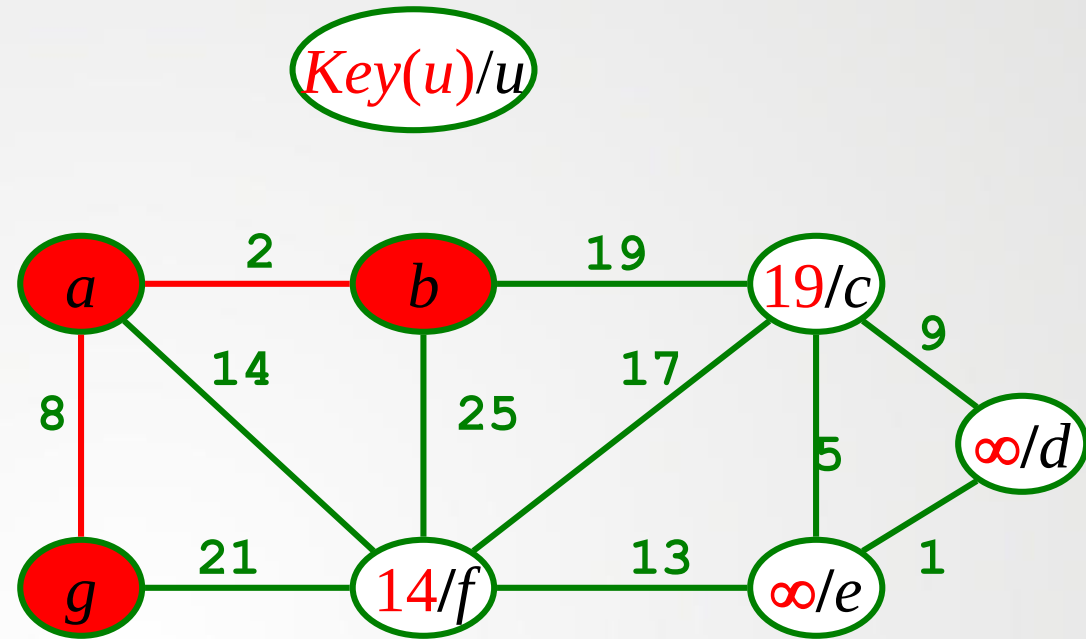
- 1 **for** each $u \in V$ **do**
- 2 $key[u] \leftarrow \infty$
- 3 $\pi[u] \leftarrow \text{NIL}$
- 4 $key[r] \leftarrow 0$
- 5 $Q \leftarrow V$
- 6 **while** $Q \neq \emptyset$ **do**
- 7 $u \leftarrow \text{ExtractMIN}(Q)$
- 8 **for** each $v \in \text{Adj}[u]$ **do**
- 9 **if** $v \in Q$ and $w(u, v) < key[v]$ **then**
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An example

MSTPrim(G, w, r)

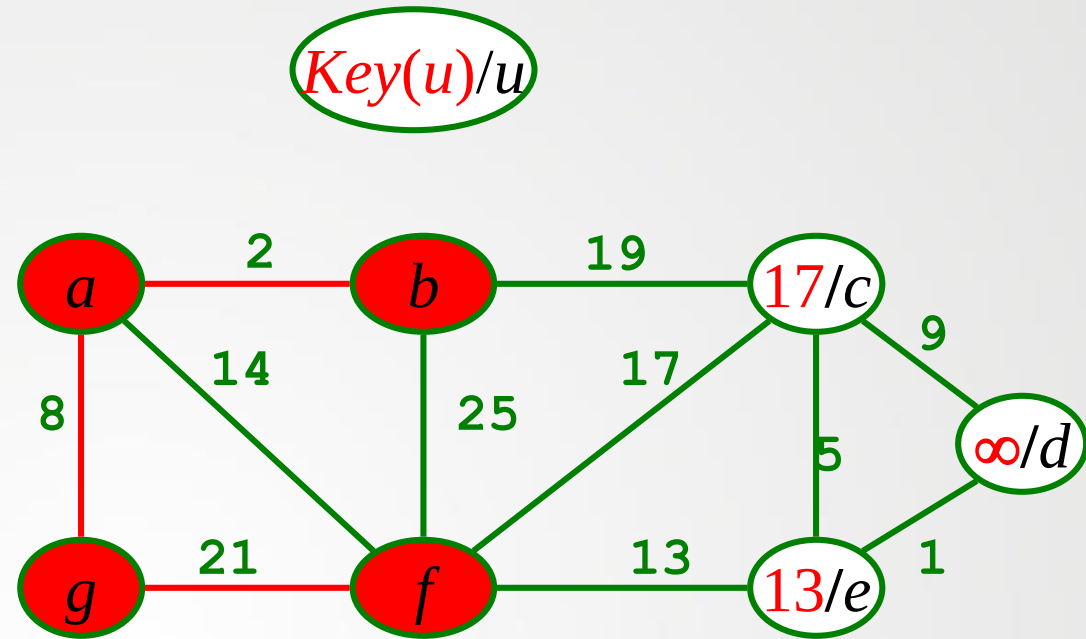
- 1 **for** each $u \in V$ **do**
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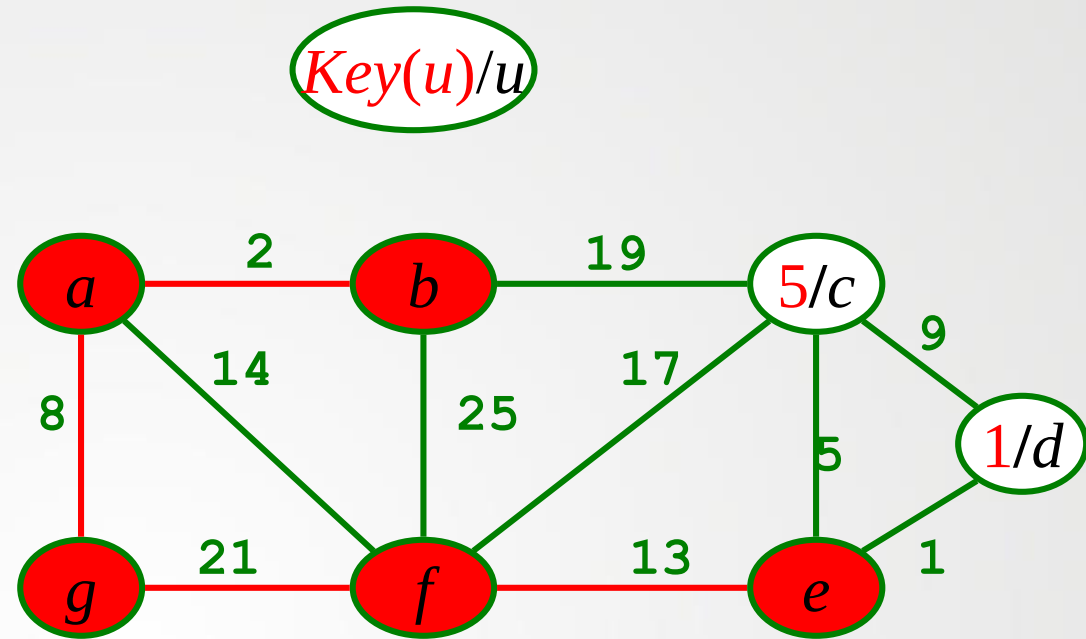
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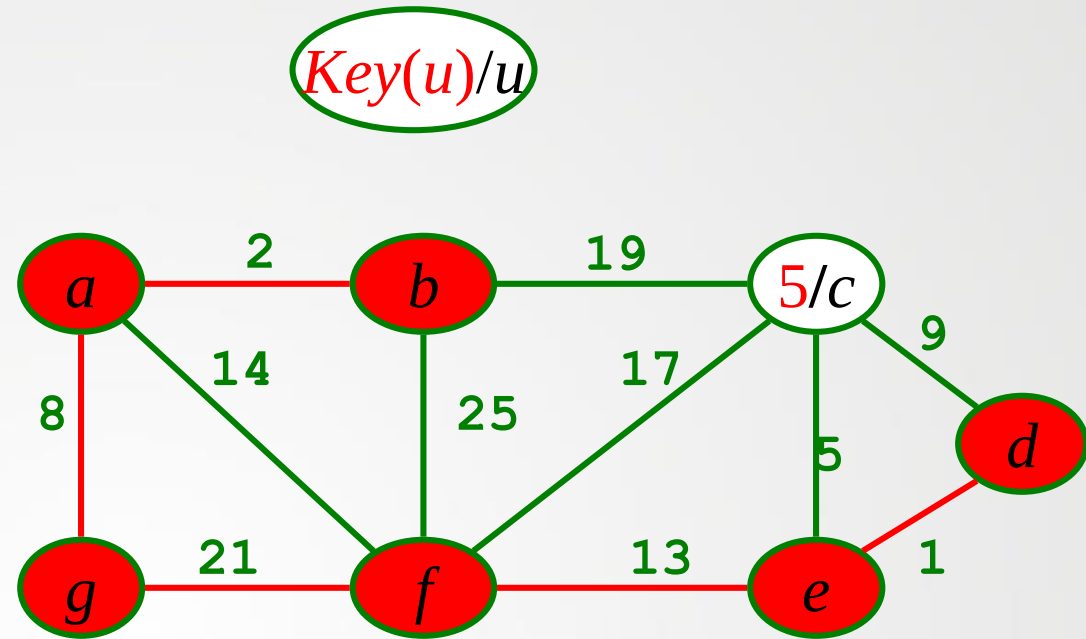
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An example

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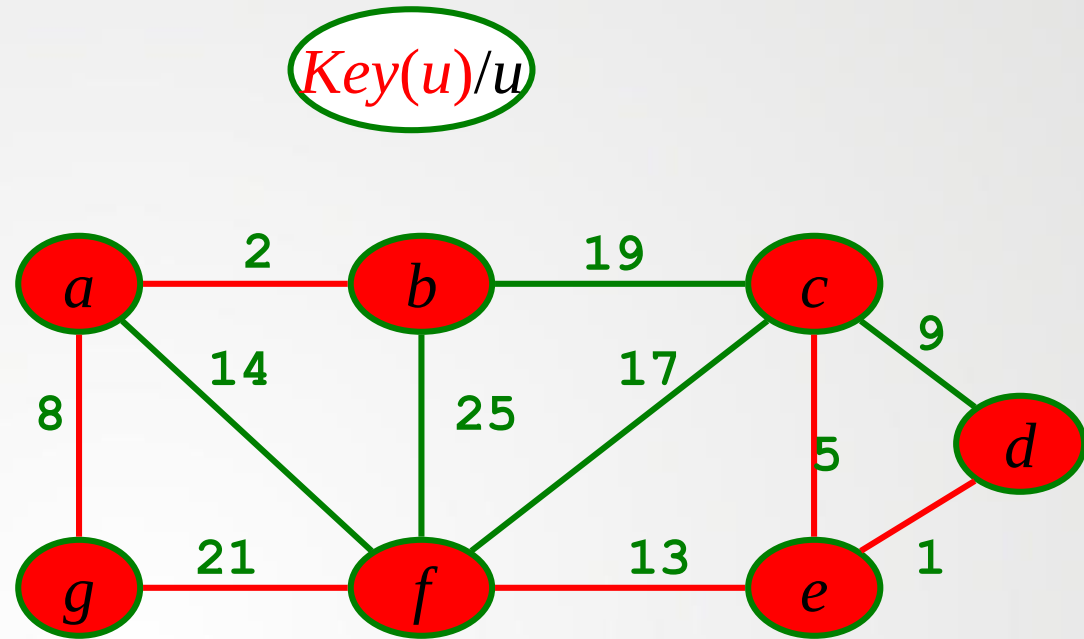
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An example

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作业

- 实现Prim算法

谢谢大家